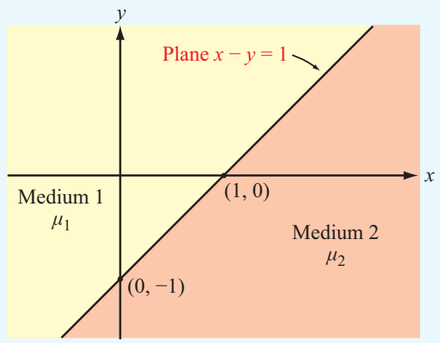


Figure P5.34 Magnetic media separated by the plane $x - y = 1$ (Problem 5.34).

5.36 Show that, if no surface current densities exist at the parallel interfaces shown in **Fig. P5.36**, the relationship between θ_4 and θ_1 is independent of μ_2.

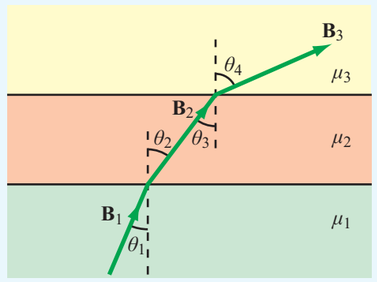


Figure P5.36 Three magnetic media with parallel interfaces (Problem 5.36).

Sections 5-7 and 5-8: Inductance and Magnetic Energy

*5.37 Obtain an expression for the self-inductance per unit length for the parallel wire transmission line of **Fig. 5-28(a)** in terms of a, d, and μ, where a is the radius of the wires, d is the axis-to-axis distance between the wires, and μ is the permeability of the medium in which they reside.

5.38 A solenoid with a length of 20 cm and a radius of 5 cm consists of 400 turns and carries a current of 12 A. If $z = 0$ represents the midpoint of the solenoid, generate a plot for $|\mathbf{H}(z)|$ as a function of z along the axis of the solenoid for the range -20 cm $\leq z \leq 20$ cm in 1 cm steps.

5.39 In terms of the dc current I, how much magnetic energy is stored in the insulating medium of a 3-m long, air-filled section of a coaxial transmission line given that the radius of the inner conductor is 5 cm and the inner radius of the outer conductor is 10 cm?

*5.40 Determine the mutual inductance between the circular loop and the linear current shown in **Fig. P5.40**.

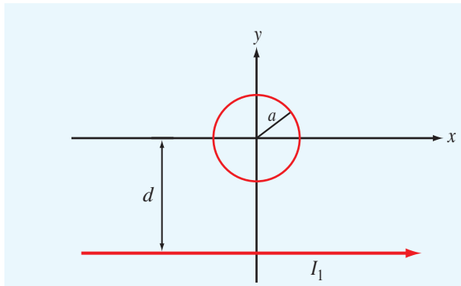


Figure P5.40 Linear conductor with current I_1 next to a circular loop of radius a at distance d (Problem 5.40).

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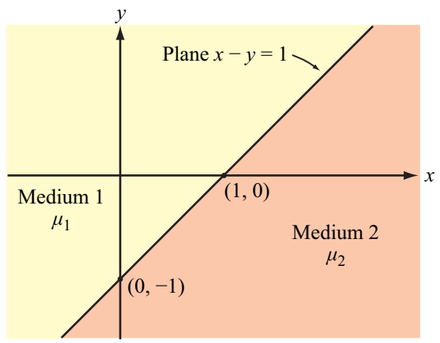


图 P5.34 由平面 x−y = 1 分隔的磁性介质（问题 5.34）。

5.36 证明，如果图 P5.36 所示的平行界面处不存在表面电流密度，则 θ_4 和 θ_1 之间的关系与 μ_2 无关。

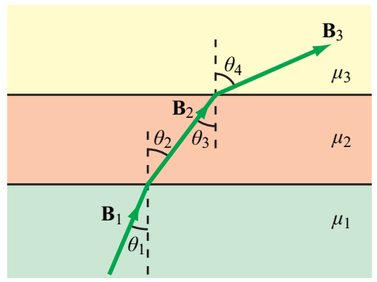


图 P5.36 具有并行接口的三个磁性介质（问题 5.36）。

第 5-7 节和 5-8 节：电感和磁能

*5.37 求出图 5-28（a）中平行导线传输线单位长度自感的表达式，用 a、d 和 μ 表示，其中 a 是导线的半径，d 是导线之间的轴距，μ 是导线所在介质的磁导率。

5.38 长度为 20 厘米、半径为 5 厘米的螺线管由 400 匝组成，电流为 12 A。如果 z = 0 代表螺线管的中点，则在 −20 cm ≤ z ≤ 20 cm 范围内，生成 |H(z)| 与沿螺线管轴 z 的函数关系图以 1 厘米为步长。

5.39 就直流电流 I 而言，假设内导体的半径为 5 cm，外导体的内半径为 10 cm，同轴传输线的 3 米长的充气部分的绝缘介质中存储了多少磁能?

*5.40 确定图P5.40所示的环形回路和线性电流之间的互感。

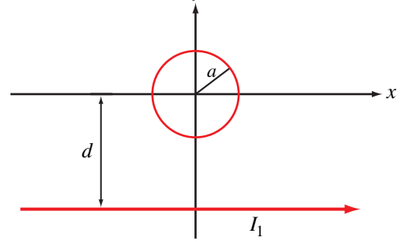
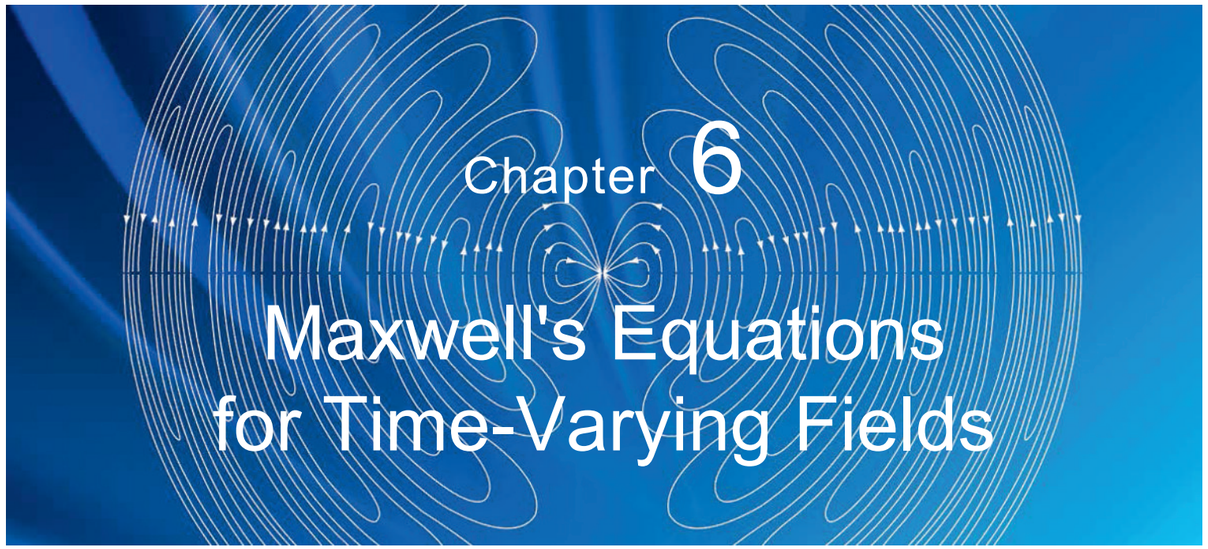


图 P5.40 电流为 I_1 的线性导体紧邻半径为 a 且距离为 d 的圆环（问题 5.40）。



Chapter 6
Maxwell's Equations for Time-Varying Fields

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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Apply Faraday's law to compute the voltage induced by a stationary coil placed in a time-varying magnetic field or moving in a medium containing a magnetic field.
2. Describe the operation of the electromagnetic generator.
3. Calculate the displacement current associated with a time-varying electric field.
4. Calculate the rate at which charge dissipates in a material with known ε and σ.

第6章

时变场的麦克斯韦方程组

章节内容

动态场，273 6-1 法拉第定律，273 6-2 时变磁场中的静止环路，274 6-3 理想变压器，278 6-4 静磁场中的移动导体，279 6-5 电磁发电机，283 6-6 时变磁场中的移动导体，284 6-7 位移电流，285 6-8 电磁学边界条件，287 6-9 电荷电流连续性关系，288 6-10 导体中的自由电荷耗散，289 TB12 EMF 传感器，290 6-11 电磁势，292 第 6 章摘要，295 问题，296

目标

学习本章介绍的材料后，您应该能够：

1. 应用法拉第定律计算放置在时变磁场中或在包含磁场的介质中移动的固定线圈所感应的电压。 2. 描述电磁发电机的工作原理。 3. 计算与时变电场相关的位移电流。 4. 计算已知 ε 和 σ 的材料中电荷耗散的速率。

Dynamic Fields

Electric charges induce electric fields and electric currents induce magnetic fields. As long as the charge and current distributions remain constant in time, so will the fields they induce. If the charges and currents vary in time, the electric and magnetic fields vary accordingly. Additionally, however, the electric and magnetic fields become coupled and travel through space in the form of electromagnetic waves. Examples of such waves include light, x-rays, infrared, gamma rays, and radio waves (see **Fig. 1-16**).

To study time-varying electromagnetic phenomena, we need to consider the entire set of Maxwell's equations simultaneously. These equations, first introduced in the opening section of Chapter 4, are given in both differential and integral form in **Table 6-1**. In the static case ($\partial/\partial t = 0$), we use the first pair of Maxwell's equations to study electric phenomena (Chapter 4) and the second pair to study magnetic phenomena (Chapter 5). In the dynamic case ($\partial/\partial t \neq 0$), the coupling that exists between the electric and magnetic fields, as expressed by the second and fourth equations in **Table 6-1**, prevents such decomposition. The first equation represents Gauss's law for electricity, and it is equally valid for static and dynamic fields. The same is true for the third equation, which is Gauss's law for magnetism. By contrast, the second and fourth equations—Faraday's and Ampère's laws—are of a totally different nature. Faraday's law expresses the fact that a time-varying magnetic field gives rise to an electric field. Conversely, Ampère's law states that a time-varying electric field must be accompanied by a magnetic field.

Some statements in this and succeeding chapters contradict conclusions reached in Chapter 4 and 5 as those pertained to the special case of static charges and dc currents. The behavior of dynamic fields reduces to that of static ones when $\partial/\partial t$ is set to zero.

We begin this chapter by examining Faraday's and Ampère's laws and some of their practical applications. We then combine Maxwell's equations to obtain relations among the charge and current sources, ρ_v and $\mathbf{J}$; the scalar and vector potentials, V and $\mathbf{A}$; and the electromagnetic fields, $\mathbf{E}$, $\mathbf{D}$, $\mathbf{H}$, and $\mathbf{B}$. We do so for the most general time-varying case and for the specific case of sinusoidal-time variations.

6-1 Faraday's Law

The close connection between electricity and magnetism was established by Oersted, who demonstrated that a wire carrying an electric current exerts a force on a compass needle and that the needle always turns to point in the $\hat{\boldsymbol{\phi}}$ direction when the current is along the $\hat{\mathbf{z}}$ direction. The force acting on the compass needle is due to the magnetic field produced by the current in the wire. Following this discovery, *Faraday hypothesized that if a current produces a magnetic field, then the converse should also be true: A magnetic field should produce a current in a wire*. To test his hypothesis, he conducted numerous experiments in his laboratory in London over a period of about 10 years—all aimed at making magnetic fields induce currents in wires. Similar work was being carried out by Henry in Albany, New York. Wires were placed next to permanent magnets or current-carrying loops of all different sizes, but no

Table 6-1 Maxwell's equations.

Reference	Differential Form	Integral Form	
Gauss's law	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	(6.1)
Faraday's law	$\nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\int_S \dfrac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$	(6.2)*
No magnetic charges (Gauss's law for magnetism)	$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	(6.3)
Ampère's law	$\nabla \times \mathbf{H} = \mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t}\right) \cdot d\mathbf{s}$	(6.4)

*For a stationary surface S.

时变场

电荷产生电场，电流产生磁场。只要电荷和电流的分布不随时间变化，它们产生的场也不随时间变化。若电荷和电流随时间变化，电场和磁场也相应变化；同时，电场与磁场相互耦合，并可作为电磁波在空间中传播。光、X 射线、红外线、γ 射线及无线电波均属电磁波（见图 1-16）。

研究时变电磁现象时，必须同时考虑完整的麦克斯韦方程组。这些方程已在第4章开头引入，其微分形式和积分形式列于表 6-1。静态情况下（∂/∂t = 0），可用前两个方程研究电现象（第4章），用后两个方程研究磁现象（第5章）。在时变情况下（∂/∂t ≠ 0），表中第二、第四个方程所描述的电磁耦合使这种分开处理不再适用。第一个方程为电场的高斯定律，对静态场和时变场均成立；第三个方程为磁场的高斯定律，同样如此。第二、第四个方程分别为法拉第定律和安培–麦克斯韦定律，其性质有所不同：法拉第定律表明时变磁场产生电场；安培–麦克斯韦定律则表明时变电场必然伴随磁场。

本章及后续章节中的某些论述看似与第4、5章的结论不同，是因为那些结论仅适用于静止电荷和直流电流。当令 ∂/∂t = 0 时，时变场的关系便退化为静态场的关系。

本章首先讨论法拉第定律、安培–麦克斯韦定律及其应用；随后联立麦克斯韦方程，建立电荷、电流源 ρ_v、J，标量势 V、矢量势 A，以及电磁场 E、D、H、B 之间的关系。讨论既包括一般的时变情况，也包括随时间作正弦变化的特殊情况。

6-1 法拉第定律

奥斯特揭示了电与磁的密切联系：载流导线对磁针施加作用力；当电流沿 +z 方向时，磁针转向方位角 φ 增大的方向。磁针受到的作用来自导线中电流产生的磁场。随后，法拉第提出假设：既然电流能够产生磁场，反过来磁场是否也能在导线中产生电流？为检验这一假设，他在伦敦实验室进行了约十年的大量实验，试图利用磁场在导线中产生感应电流。同一时期，亨利也在纽约州奥尔巴尼开展类似研究。他们将导线放在永久磁铁或各种尺寸的载流回路旁边，但并未……

（下页续）

表 6-1 麦克斯韦方程组

定律	微分形式	积分形式	
电场的高斯定律	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	(6.1)
法拉第定律	$\nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\int_S \dfrac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$	(6.2)*
不存在磁荷（磁场的高斯定律）	$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	(6.3)
安培–麦克斯韦定律	$\nabla \times \mathbf{H} = \mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \dfrac{\partial \mathbf{D}}{\partial t}\right) \cdot d\mathbf{s}$	(6.4)

* 适用于静止曲面 S。

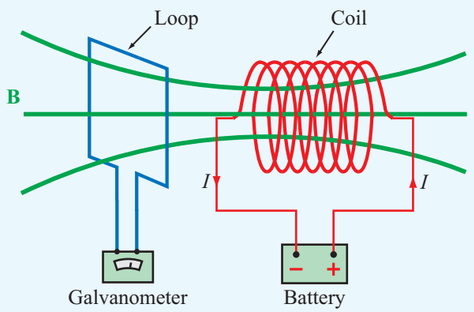


Figure 6-1 The galvanometer (predecessor of the ammeter) shows a deflection whenever the magnetic flux passing through the square loop changes with time.

currents were ever detected. Eventually, these experiments led to the following discovery by both Faraday and Henry.

► Magnetic fields can produce an electric current in a closed loop, but *only* if the magnetic flux linking the surface area of the loop changes with time. The key to the induction process is *change*. ◄

To elucidate the induction process, consider the arrangement shown in **Fig. 6-1**. A conducting loop connected to a galvanometer (a sensitive instrument used in the 1800s to detect current flow) is placed next to a conducting coil connected to a battery. The current in the coil produces a magnetic field **B** whose lines pass through the loop. In Section 5-4, we defined the magnetic flux Φ passing through a loop as the integral of the normal component of the magnetic flux density over the surface area of the loop, S, or

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \qquad \text{(Wb)}. \tag{6.5}$$

Under stationary conditions, the dc current in the coil produces a constant magnetic field **B**, which in turn produces a constant flux through the loop. When the flux is constant, no current is detected by the galvanometer. However, when the battery is disconnected, thereby interrupting the flow of current in the coil, the magnetic field drops to zero, and the consequent change in magnetic flux causes a momentary deflection of the galvanometer needle. When the battery is reconnected, the galvanometer again exhibits a momentary deflection, but in the opposite direction. Thus, current is induced in the loop when the magnetic flux changes, and the direction of the current depends on whether the flux increases (when the battery is being connected) or decreases (when the battery is being disconnected). It was further discovered that current can also flow in the loop while the battery is connected to the coil if the loop turns or moves either closer to or away from the coil. The physical movement of the loop changes the amount of flux linking its surface S, even though the field **B** due to the coil has not changed.

A galvanometer is a predecessor of the voltmeter and ammeter. When a galvanometer detects the flow of current through the coil, it means that a voltage has been induced across the galvanometer terminals. This voltage is called the ***electromotive force*** (***emf***), V_{emf}, and the process is called ***electromagnetic induction***. The emf induced in a closed conducting loop of N turns is given by

$$V_{\text{emf}} = -N\frac{d\Phi}{dt} = -N\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{s} \qquad \text{(V)}. \tag{6.6}$$

Even though the results leading to Eq. (6.6) were also discovered independently by Henry, Eq. (6.6) is attributed to Faraday and known as ***Faraday's law***. The significance of the negative sign in Eq. (6.6) is explained in the next section.

We note that the derivative in Eq. (6.6) is a total time derivative that operates on the magnetic field **B**, as well as the differential surface area $d\mathbf{s}$. Accordingly, an emf can be generated in a closed conducting loop under any of the following three conditions:

1. A time-varying magnetic field linking a stationary loop; the induced emf is then called the ***transformer emf***, $V_{\text{emf}}^{\text{tr}}$.
2. A moving loop with a time-varying area (relative to the normal component of **B**) in a static field **B**; the induced emf is then called the ***motional emf***, $V_{\text{emf}}^{\text{m}}$.
3. A moving loop in a time-varying field **B**.

The total emf is given by

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} \tag{6.7}$$

with $V_{\text{emf}}^{\text{m}} = 0$ if the loop is stationary [case (1)] and $V_{\text{emf}}^{\text{tr}} = 0$ if **B** is static [case (2)]. For case (3), both terms are important. Each of the three cases is examined separately in the following sections.

6-2 Stationary Loop in a Time-Varying Magnetic Field

The stationary, single-turn, conducting, circular loop with contour C and surface area S shown in **Fig. 6-2(a)** is exposed

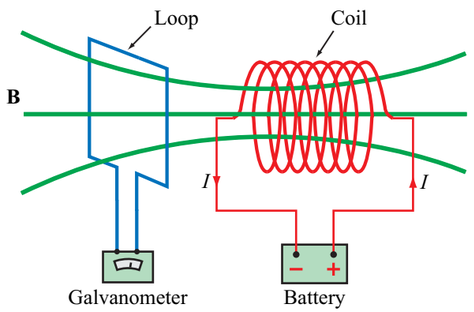


图 6-1 每当穿过方形环路的磁通量随时间变化时，检流计（电流表的前身）就会显示出偏转。

曾经检测到电流。最终，这些实验导致了法拉第和亨利的以下发现。

► 磁场可以在闭环中产生电流，但前提是连接环路表面区域的磁通量随时间变化。归纳过程的关键是改变。 ◄

为了阐明归纳过程，请考虑图 6-1 中所示的布置。连接到检流计（一种 1800 年代用于检测电流的灵敏仪器）的导电环被放置在连接到电池的导电线圈旁边。线圈中的电流产生磁场 B，其磁力线穿过环路。在第 5-4 节中，我们将穿过环路的磁通量 Φ 定义为环路表面积上磁通密度法向分量 S 的积分，或

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \qquad \text{(Wb)}. \tag{6.5}$$

在静止条件下，线圈中的直流电流产生恒定磁场 B，进而产生通过环路的恒定磁通。当通量恒定时，检流计检测不到电流。然而，当电池断开时，线圈中的电流中断，磁场降至零，磁通量随之变化，导致检流计针瞬时偏转。当电池重新连接时，检流计再次表现出瞬时偏转，但在相反的方向。因此，当磁通量变化时，回路中会感应出电流，电流的方向取决于磁通量是增加（连接电池时）还是减少（当电池断开时）。进一步发现，当电池连接到线圈时，如果环路转动或靠近或远离线圈，电流也会在环路中流动。环路的物理运动改变了连接其表面 S 的磁通量，尽管线圈产生的磁场 B 并未改变。

检流计是电压表和电流表的前身。当检流计检测到流过线圈的电流时，这意味着检流计端子上已感应出电压。该电压称为电动势 (emf) V_{emf}，该过程称为电磁感应。N 匝闭合导电回路中感应的电动势由下式给出

$$V_{\text{emf}} = -N\frac{d\Phi}{dt} = -N\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{s} \qquad \text{(V)}. \tag{6.6}$$

即使结果导致方程。 (6.6) 也是由亨利独立发现的，方程。 (6.6) 归因于法拉第，被称为法拉第定律。方程中负号的意义。 (6.6) 将在下一节中解释。

我们注意到方程中的导数。 (6.6) 是作用于磁场 B 以及微分表面积 ds 的总时间导数。因此，在以下三个条件中的任何一个条件下，闭合导电回路中都会产生电动势：

1. 连接静止环路的时变磁场；感应电动势称为变压器电动势，$V_{\text{emf}}^{\text{tr}}$。 2. 静态场 B 中具有时变面积（相对于 B 的法向分量）的移动环路；感应电动势称为运动电动势，$V_{\text{emf}}^{\text{m}}$。 3. 时变场中的移动环 B.

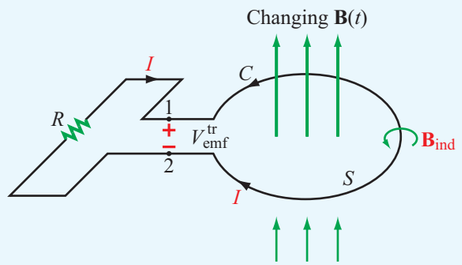
总电动势由下式给出

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}} \tag{6.7}$$

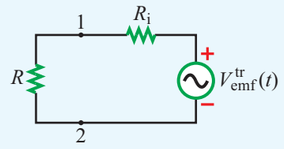
如果循环是静态的，则 $V_{\text{emf}}^{\text{m}} = 0$ [情况 (1)]；如果 B 是静态的，则 $V_{\text{emf}}^{\text{tr}} = 0$ [情况 (2)]。对于情况 (3)，这两个术语都很重要。以下各节将分别研究这三种情况。

6-2 时变磁场中的静止环路

暴露出如图 6-2(a) 所示的具有轮廓 C 和表面积 S 的固定单匝导电圆形回路



(a) Loop in a changing **B** field



(b) Equivalent circuit

Figure 6-2 (a) Stationary circular loop in a changing magnetic field $\mathbf{B}(t)$ and (b) its equivalent circuit.

to a time-varying magnetic field $\mathbf{B}(t)$. As stated earlier, the emf induced when S is stationary and the field is time varying is called the ***transformer emf*** and is denoted $V_{\text{emf}}^{\text{tr}}$. Since the loop is stationary, d/dt in Eq. (6.6) now operates on $\mathbf{B}(t)$ only. Hence,

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad \textbf{(transformer emf)} \tag{6.8}$$

where the full derivative d/dt has been moved inside the integral and changed into a partial derivative $\partial/\partial t$ to signify that it operates on **B** only. The transformer emf is the voltage difference that would appear across the small opening between terminals 1 and 2, even in the absence of the resistor R. That is, $V_{\text{emf}}^{\text{tr}} = V_{12}$, where V_{12} is the open-circuit voltage across the open ends of the loop. Under dc conditions, $V_{\text{emf}}^{\text{tr}} = 0$. For the loop shown in **Fig. 6-2(a)** and the associated definition for $V_{\text{emf}}^{\text{tr}}$ given by Eq. (6.8), the direction of $d\mathbf{s}$, which is the loop's differential surface normal, can be chosen to be either upward or downward. The two choices are associated with opposite designations of the polarities of terminals 1 and 2 in **Fig. 6-2(a)**.

Right-Hand Rule for V_{emf}

The connection between the direction of $d\mathbf{s}$ and the polarity of $V_{\text{emf}}^{\text{tr}}$ is governed by the following right-hand rule: If $d\mathbf{s}$ points along the thumb of the right hand, then the direction of the contour C indicated by the four fingers is such that it always passes across the opening from the positive terminal of $V_{\text{emf}}^{\text{tr}}$ to the negative terminal.

If the loop has an internal resistance R_{i}, the circuit in **Fig. 6-2(a)** can be represented by the equivalent circuit shown in **Fig. 6-2(b)**, in which case the current I flowing through the circuit is given by

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_{\text{i}}}. \tag{6.9}$$

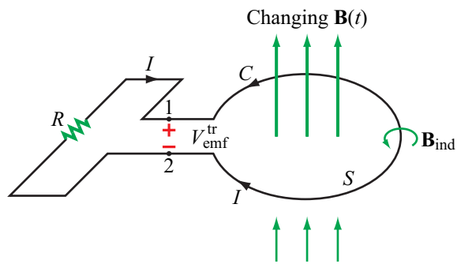
For good conductors, R_{i} usually is very small, and it may be ignored in comparison with practical values of R.

► The polarity of $V_{\text{emf}}^{\text{tr}}$ and hence the direction of I is governed by ***Lenz's law***, which states that the current in the loop is always in a direction that opposes the change of magnetic flux $\Phi(t)$ that produced I. ◄

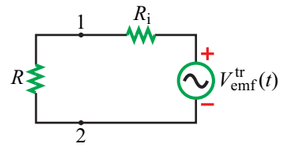
The current I induces a magnetic field of its own, $\mathbf{B}_{\text{ind}}$, with a corresponding flux Φ_{ind}. The direction of $\mathbf{B}_{\text{ind}}$ is governed by the right-hand rule: If I is in a clockwise direction, then $\mathbf{B}_{\text{ind}}$ points downward through S; conversely, if I is in a counterclockwise direction, then $\mathbf{B}_{\text{ind}}$ points upward through S. If the original field $\mathbf{B}(t)$ is increasing, which means that $d\Phi/dt > 0$, then according to Lenz's law I has to be in the direction shown in **Fig. 6-2(a)** in order for $\mathbf{B}_{\text{ind}}$ to be in opposition to $\mathbf{B}(t)$. Consequently, terminal 2 would be at a higher potential than terminal 1, and $V_{\text{emf}}^{\text{tr}}$ would have a negative value. However, if $\mathbf{B}(t)$ were to remain in the same direction but to decrease in magnitude, then $d\Phi/dt$ would become negative, the current would have to reverse direction, and its induced field $\mathbf{B}_{\text{ind}}$ would be in the same direction as $\mathbf{B}(t)$ in order to oppose the change (decrease) of $\mathbf{B}(t)$. In that case, $V_{\text{emf}}^{\text{tr}}$ would be positive.

► It is important to remember that $\mathbf{B}_{\text{ind}}$ serves to oppose the *change* in $\mathbf{B}(t)$ and not necessarily $\mathbf{B}(t)$ itself. ◄

Despite the presence of the small opening between terminals 1 and 2 of the loop in **Fig. 6-2(a)**, we shall treat the loop as a closed path with contour C. We do this in order to establish the link between **B** and the electric field **E** associated with the induced emf, $V_{\text{emf}}^{\text{tr}}$. Also, at any point along the loop, the



(a) Loop in a changing **B** field



(b) Equivalent circuit

图 6-2 (a) 变化磁场 B(t) 中的静止圆环和 (b) 其等效电路。

为时变磁场 B(t)。如前所述，当 S 静止并且磁场随时间变化时感应的电动势称为变压器电动势，并表示为 $V_{\text{emf}}^{\text{tr}}$。由于循环是平稳的，式中的 d/dt (6.6) 现在仅适用于 B(t)。因此，

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \quad \textbf{(transformer emf)} \tag{6.8}$$

其中全导数 d/dt 已移至积分内部并更改为偏导数 ∂/∂t 以表示它仅对 B 进行运算。变压器电动势是端子 1 和 2 之间的小开口上出现的电压差，即使没有电阻器 R。即 $V_{\text{emf}}^{\text{tr}} = V_{12}$，其中 V_{12} 是环路开路端的开路电压。在直流条件下，$V_{\text{emf}}^{\text{tr}} = 0$。对于图 6-2(a) 中所示的循环以及由方程给出的 $V_{\text{emf}}^{\text{tr}}$ 的相关定义：(6.8) 中，ds 的方向是环的微分面法线，可以选择向上或向下。这两种选择与图 6-2(a) 中端子 1 和 2 极性的相反指定相关。

V_{emf} 的右手定则

ds 的方向和 $V_{\text{emf}}^{\text{tr}}$ 的极性之间的关系受以下右手定则约束：如果 ds 沿着右手的拇指指向，则四个手指所指示的轮廓 C 的方向总是穿过从 $V_{\text{emf}}^{\text{tr}}$ 的正极端子到负极端子的开口。

如果回路有一个内阻R_{i}，图6-2(a)中的电路可以用图6-2(b)所示的等效电路表示，此时流过电路的电流I为

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_{\text{i}}}. \tag{6.9}$$

对于良导体，R_{i}通常很小，与R的实际值相比可以忽略不计。

► $V_{\text{emf}}^{\text{tr}}$ 的极性以及 I 的方向受楞次定律控制，该定律规定环路中的电流始终处于与产生 I 的磁通量 Φ(t) 变化相反的方向。◄

电流 I 感应出自己的磁场 B_{ind}，并具有相应的磁通量 Φ_{ind}。B_{ind}的方向受右手定则约束：如果I沿顺时针方向，则B_{ind}通过S指向向下；反之，如果 I 沿逆时针方向，则 B_{ind} 通过 S 指向上方。如果原始场 B(t) 递增，即 dΦ/dt > 为 0，则根据楞次定律，I 必须沿图 6-2(a) 所示方向，B_{ind} 才能与 B(t) 相反。因此，端子 2 的电位将高于端子 1，并且 $V_{\text{emf}}^{\text{tr}}$ 将具有负值。然而，如果 B(t) 保持相同方向但幅度减小，则 dΦ/dt 将变为负值，电流必须反转方向，并且其感应场 B_{ind} 将与 B(t) 方向相同，以抵抗 B(t) 的变化（减小）。在这种情况下，$V_{\text{emf}}^{\text{tr}}$ 将为正。

► 重要的是要记住，B_{ind} 用于反对 B(t) 的变化，而不一定是 B(t) 本身。◄

尽管图 6-2(a) 中环路的端子 1 和 2 之间存在小开口，但我们仍应将环路视为具有轮廓 C 的闭合路径。这样做是为了在 B 和与感应电动势 $V_{\text{emf}}^{\text{tr}}$ 相关的电场 E 之间建立联系。此外，在循环中的任何一点，